# Ashwin Deshpande Econometrician

**Quantitative data scientist** with 4 years of experience applying **Applied Time Series** and **Structural Econometrics** to solve complex financial and market-design problems. Expert in **causal inference**, regime-detection, and high-dimensional data analysis. Proven track record of translating econometric theory into production-scale automated systems using **Python, R, and SQL** to guide multi-million dollar portfolio and policy decisions.

## EDUCATION

University of Illinois, Chicago — *M.S. in Computer Science - 4.0 -* 2022
B.V.B College of Engineering & Technology, Hubli — *B.Eng in Computer Science -* 2018

## CONTACT


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**Github**: https://github.com/AshwinDeshpande96 **Technical blog:** https://ashwindeshpande96.github.io/personal_webpage/


## SKILLS

**Econometric Theory:** Time Series (ARIMA, Prophet, Bayesian VAR, GARCH), X-13ARIMA-SEATS/STL, Causal Inference, Cointegration, Panel Data, Structural Break Detection

**Inference Tools:** R (Advanced), Python (Statsmodels, Scikit-learn, PyTorch), SQL, Matlab.

**Systems & Platforms:** AWS, GCP, BigQuery, Airflow, MLflow, Docker, Vertex AI, Google TimesFM

**Business Domain:** Asset Pricing, Macro-Forecasting, Market Microstructure, Scenario Analysis.

## PROFESSIONAL EXPERIENCE

Thynk Capital LLC (AI Hedge Fund) — Senior Data Scientist (Econometrics & Strategy) | *May 2023 – Feb 2025*

- Developed **alpha-generating structural models** using multivariate time-series and cointegration analysis to isolate causal drivers across asset classes.
- Built forecasting frameworks incorporating **structural breaks and macroeconomic drivers** to evaluate portfolio sensitivity under synthetic "shock" scenarios.
- Productionized econometrics-based models using **Python and Airflow**, ensuring high-frequency adaptability to shifting market regimes.
- Partnered with leadership to translate **high-dimensional statistical variance** into actionable portfolio construction and asset pricing strategy.

Reserve Bank of India — Research Intern (Statistical Modeling & Macro-Policy) | *Jan 2026 – Present*

- Built **structural VECM/BVAR models** to analyze commodities price transmission and supply-side shocks to regions & India's Consumer Price Index Inflation.
- Applied **regime-switching models** and structural break detection to identify shocks and their implications - magnitude, persistence & recurrence.
- Performed causal tests to attribute source of shocks using **difference-in-differences**, synthetic control & regression analysis methods
- Utilized Skewed Student-t distributions and multivariate **GARCH**-based frameworks to quantify **tail-risk** and communicate variance drivers to senior policy stakeholders.
- Developed spatio-temporal dashboards to visualize shock spillovers & cointegrations between regions using **GFEVD/GIRF/Johansen** for key macro-variables (Inflation, Quantity, Price).

Bombora — Data Scientist (Predictive Analytics) - Remote, USA | *Jun 2022 – Dec 2022*

- Built ML pipelines in **Vertex AI** to process millions of profiles; optimized targeting models that reduced false positives by 8%, improving the efficiency of resource allocation.
- Developed Tableau/Plotly dashboards to explain model variance and performance metrics to cross-functional partners.

UIC NLP Lab — Graduate Research Assistant - Chicago, IL - Jan 2021 – May 2022

- Built speech-to-text ML pipelines and fine-tuned **BERT**-based NLP models for dementia detection. Published findings in **ACL BioNLP 2022**: https://aclanthology.org/2022.bionlp-1.4/

Primenumbers Technologies — Data Science Engineer - Bangalore, India - May 2019 – May 2020

- Conducted A/B testing and statistical inference to measure business impact of model deployment.
- Engineered **ETL** pipelines to ingest & clean raw government contract data using Python/**PostgreSQL**

## PROJECTS

- E-commerce rank optimization of financial products for increasing EPV [Code] [Slides]
  - *Earnings per visitor | Position bias | Inverse Propensity Weighting | Google Analytics | Click-through-rate*
- Survival Analysis: Time-to-event analysis using hierarchical mixture models [Blog]
  - *R | Bayesian Inference | Survival Model | Non-parametric Hazard Model | Kaplan Meier*

## CERTIFICATES

Linear Algebra Bayesian Statistics - L1 Bayesian Statistics - L2 Deep Learning.ai-Intro to MLOps Deeplearning.ai-ML Data Lifecycle
FRM Level I Candidate (Nov 2026)